

SIMATS ENGINEERING ASSESSMENT TOOL 2

24/20

COURSE NAME: CSA07 COURSE CODE: Computer Networks

COURSE OUTCOME COVERED

CO1: Apply the functions of OSI layer in enabling reliable data transmission across networks. (BL2, BL3)

Assessment Tool Weightage: 30%

Annexure A

CONCEPT MAPPING

Code of Conduct:

Shonika (192511387)

I Betty D. (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Betty

Annexure A

CONCEPT MAPPING

1. A partially completed concept map is given below:

Application → Presentation → _____ → Transport → Network → _____ → Physical.

Analyze the missing layers and explain how their functions are essential for reliable communication.

2. Construct a concept map linking data units (Segment, Packet, Frame, Bits) with their respective OSI layers and analyze how transformation of data units supports transmission.

3. Given the concept map:

DataLink → MACAddress → LocalDelivery

Network → IP Address → Global Delivery

Explain how these relationships ensure complete data delivery across networks.

4. A concept map shows:

ErrorControl

→ DataLinkLayer → ErrorDetection

→ Transport Layer → Error Correction

Analyze how the interaction between these layers improves reliability in data transmission.

5. Develop a concept map for web browsing using OSI layers and analyze how failure in one layer affects the entire communication process.

6. A network issue is represented in the concept map:

Physical ✓ → Data Link ✓ → Network ✗ → Transport ✗ → Application ✗

Analyze the problem and explain how the concept map helps in troubleshooting.

RUBRICS FOR CALCULATION

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Concept Identification	20%	Accurately identifies all relevant OSI layers, concepts, and relationships	Identifies most concepts with minor omissions	Identifies limited concepts; some inaccuracies	Unable to identify key concepts correctly
Concept Mapping Structure	25%	Constructs a clear, logical, and well-organized concept map with proper hierarchy and connections	Concept map is mostly clear with minor structural issues	Concept map lacks clarity, organization, or proper flow	No clear structure or incorrect mapping
Linking & Relationships	20%	Clearly establishes correct relationships between layers, functions, and processes	Relationships are mostly correct with minor errors	Limited or partially correct relationships	Incorrect or missing relationships
Application & Analysis	20%	Effectively applies concepts to explain processes (e.g., encapsulation, error control) with strong reasoning	The application is correct but lacks depth in analysis	Basic application with weak explanation	Unable to apply concepts correctly
Explanation & Clarity	15%	Provides a clear, concise, and well-structured explanation supporting the concept map	Explanation is understandable with minor gaps	Explanation is unclear or incomplete	No clear or meaningful explanation

Q1.Missing OSI Layers and Their Role in Reliable Communication

The Open Systems Interconnection (OSI) model is a seven-layer framework that enables communication between different network devices. Each layer performs a specific function and passes data to the next layer. In the given sequence,

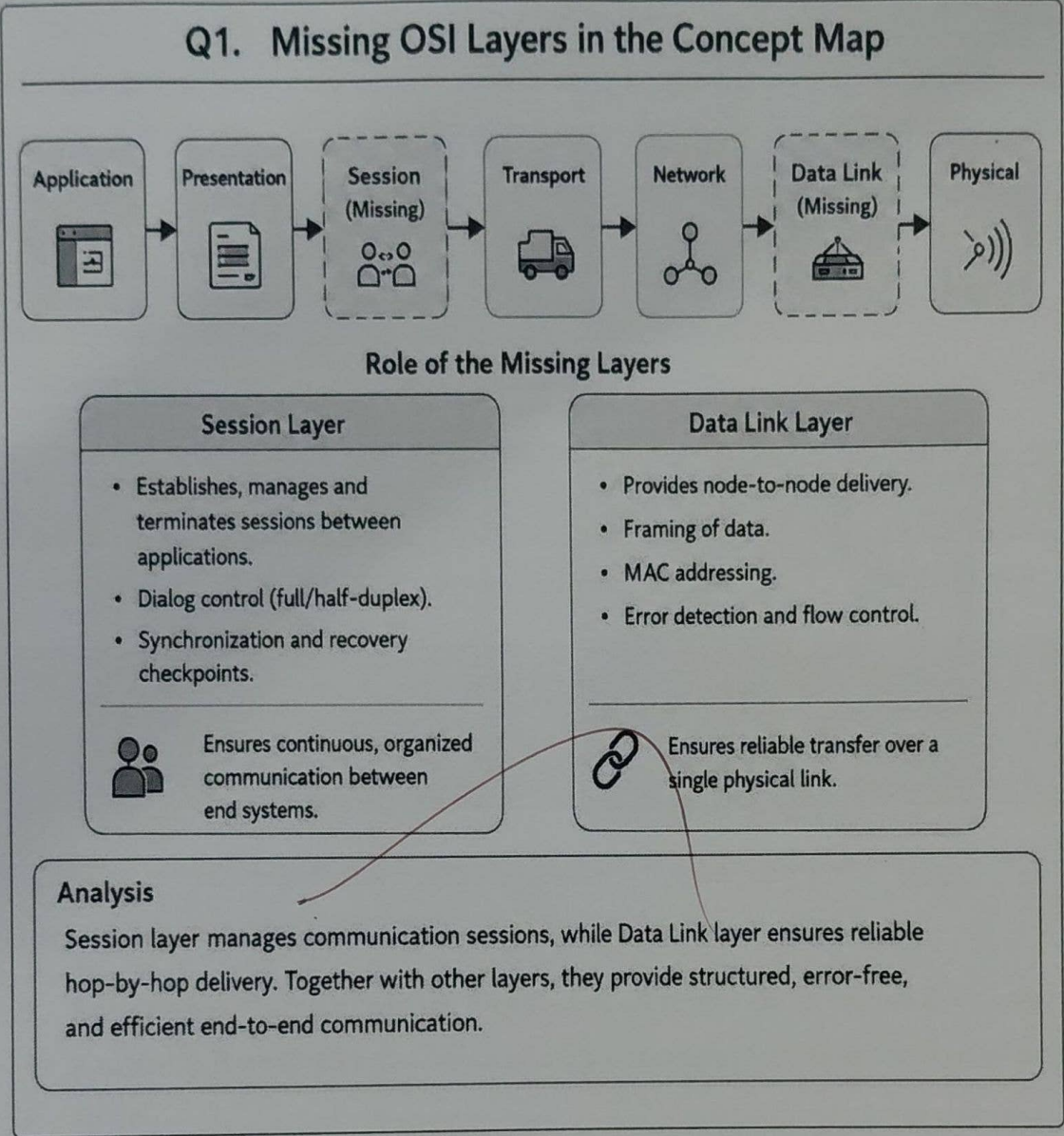
Application → Presentation → _____ → Transport → Network → _____ → Physical

the missing layers are Session Layer and Data Link Layer.

The Session Layer establishes, manages, and terminates communication sessions between applications. It ensures that data exchange remains synchronized and allows interrupted sessions to resume without losing information. This improves communication reliability, especially during long data transfers.

The Data Link Layer provides reliable ~~node-to-node~~ communication. It converts packets into frames, detects transmission errors using error detection methods, controls data flow, and manages MAC addressing. These functions help ensure that data reaches the next device without corruption.

Together, all seven OSI layers work cooperatively to provide accurate, secure, and reliable communication across networks.



Concept Map Architecture (Q1)

Source Concept (Node A)	Linking Relationship	Target Concept (Node B)
Application Layer	Sends user requests to	Presentation Layer
Presentation Layer	Formats and encrypts data for	Session Layer

Session Layer	establishes and manages	Communication Session
Communication Session	transfers data through	Transport Layer
Transport Layer	provides reliable end-to-end delivery to	Network Layer
Network Layer	routes packets through	Data Link Layer
Data Link Layer	converts packets into	Frames
Frames	transmitted over	Physical Layer
Physical Layer	carries data as	Bits through transmission media

Literature Insights & Evidence (Q1)

- **Forouzan (2022):** Explains the functions of all seven OSI layers and their interaction.
- **Kurose & Ross (2021):** Describes reliable communication using the layered network model.
- **Tanenbaum & Wetherall (2021):** Discusses session management and data link responsibilities.
- **Stallings (2020):** Highlights error detection and flow control in the Data Link Layer.
- **Cisco Networking Academy:** Explains how the OSI model supports efficient data communication.

Q2. Data Units and Their Transformation in the OSI Model

During communication, data passes through different OSI layers, where each layer adds its own control information. This process is called Encapsulation. As the data moves down the layers, its format changes from Segment to Packet, then Frame, and finally Bits for transmission.

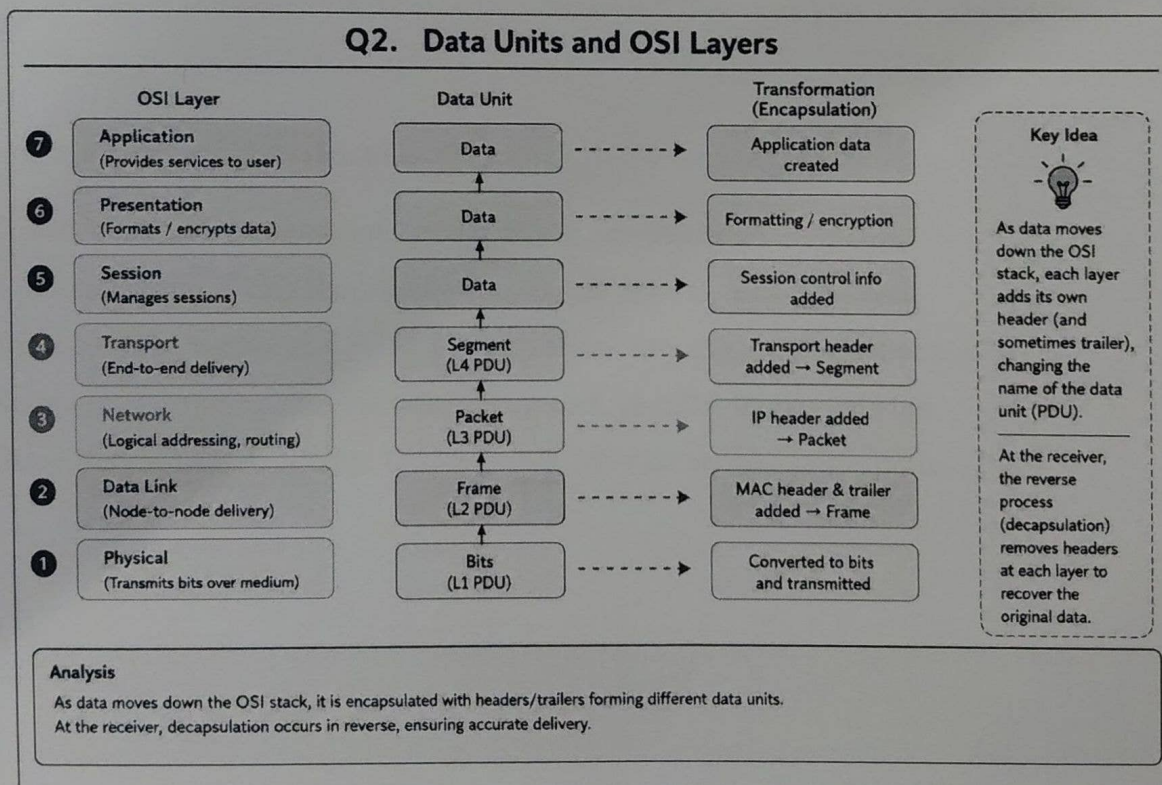
The Transport Layer divides application data into Segments and provides reliable delivery through sequencing and error recovery.

The Network Layer adds logical addressing and converts segments into Packets, allowing routing between networks.

The Data Link Layer converts packets into Frames by adding MAC addresses and error detection information, enabling communication within a local network.

Finally, the Physical Layer converts frames into Bits, which are transmitted as electrical, optical, or radio signals through the transmission medium.

At the receiving end, the reverse process (Decapsulation) removes the headers layer by layer until the original data reaches the application.



Concept Map Architecture (Q2)

Source Concept (Node A)	Linking Relationship	Target Concept (Node B)
Application Data	divided into	Segment
Segment	created by	Transport Layer
Transport Layer	passes data as	Packet
Packet	created by	Network Layer
Network Layer	converts packet into	Frame
Frame	created by	Data Link Layer
Data Link Layer	converts frame into	Bits
Bits	transmitted through	Physical Layer
Physical Layer	delivers bits to	Receiving Device
Receiving Device	performs	Decapsulation

Literature Insights & Evidence (Q2)

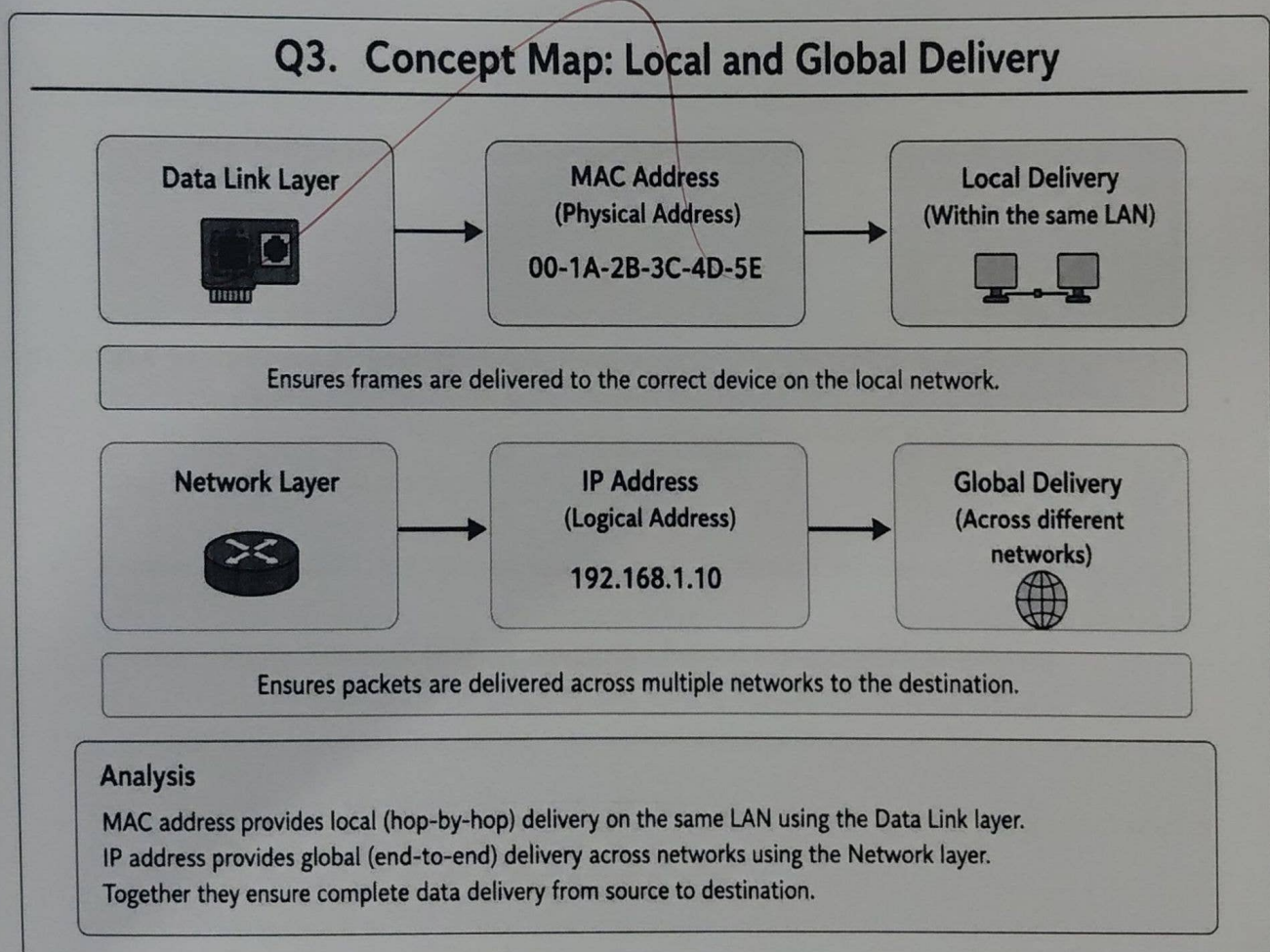
- Forouzan (2022): Explains encapsulation and decapsulation.
- Kurose & Ross (2021): Discusses protocol data units at each OSI layer.
- Tanenbaum & Wetherall (2021): Describes how data changes format across layers.
- Stallings (2020): Explains the role of segments, packets, frames, and bits.
- Cisco Networking Academy: Demonstrates OSI data encapsulation process.

Q3. Given the Concept Map: Data Link → MAC Address → Local Delivery | Network → IP Address → Global Delivery

The Data Link Layer uses MAC (Media Access Control) addresses to deliver data between devices within the same local network (LAN). Since MAC addresses are unique hardware addresses assigned to network interface cards, they ensure that frames reach the correct destination device locally.

The Network Layer uses IP (Internet Protocol) addresses to identify devices across different networks. Routers examine IP addresses to determine the best path for forwarding packets from the source network to the destination network.

Both layers work together to provide complete communication. The Network Layer ensures that data reaches the correct network, while the Data Link Layer ensures delivery to the correct device within that network. This coordination enables reliable end-to-end communication.



Concept Map Architecture (Q3)

Source Concept (Node A)	Linking Relationship	Target Concept (Node B)
Data Link Layer	Uses	MAC Address
MAC Address	Identifies	Local Device
Local Device	Enables	Local Delivery
Network Layer	Uses	IP Address
IP Address	Identifies	Destination Network
Destination Network	Enables	Global Delivery
Local Delivery	Works together with	Global Delivery
Local & Global Delivery	Ensure	Complete Data Communication

Literature Insights & Evidence (Q3)

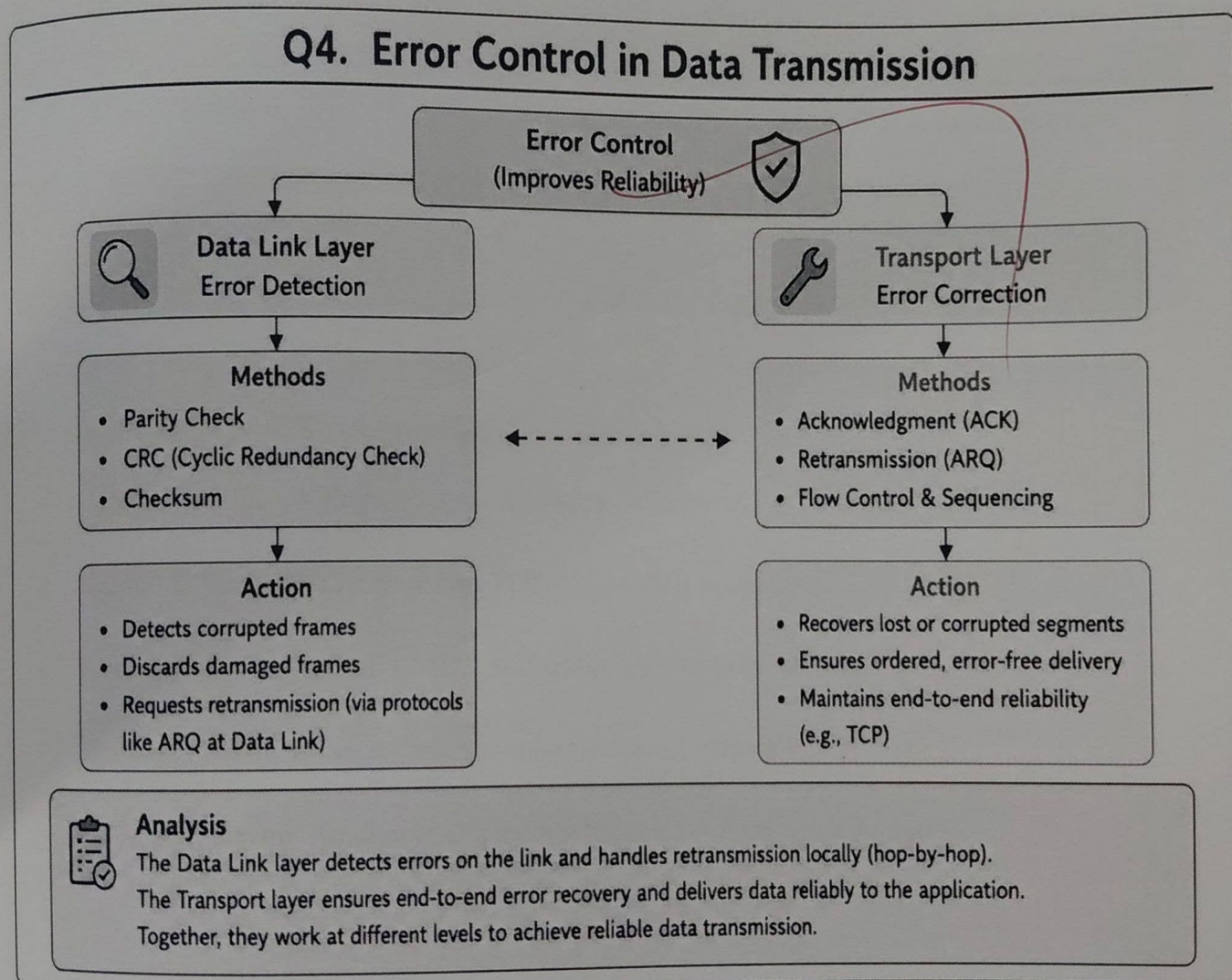
- Forouzan (2022): MAC addresses support local communication while IP addresses support inter-network communication.
- Kurose & Ross (2021): Explains addressing at different OSI layers.
- Tanenbaum & Wetherall (2021): Discusses logical and physical addressing.
- Stallings (2020): Describes the role of routers and switches.
- Cisco Networking Academy: Explains MAC and IP addressing with practical examples.

Q4. Error Control in Data Link and Transport Layers

Reliable communication requires error control at multiple layers of the OSI model. The Data Link Layer detects transmission errors between directly connected devices using techniques such as CRC (Cyclic Redundancy Check) and checksums. If an error is detected, the damaged frame is discarded or retransmitted.

The Transport Layer provides end-to-end reliability between the sender and receiver. It uses sequence numbers, acknowledgements (ACKs), timers, and retransmissions to recover lost or corrupted segments. TCP is the most common protocol providing these services.

Together, these layers improve communication reliability by detecting, correcting, and recovering from transmission errors.



Concept Map Architecture(Q4)

Source Concept (Node A)	Linking Relationship	Target Concept (Node B)
Error Control	Implemented by	Data Link Layer
Data Link Layer	Performs	Error Detection
Error Detection	Uses	CRC / Checksum
Error Control	Also Implemented by	Transport Layer
Transport Layer	Performs	Error Correction
Error Correction	Uses	ACK & Retransmission
Both Layers	Improve	Reliable Data Transmission

Literature Insights & Evidence (Q4)

- Forouzan (2022): Describes error detection techniques.
- Kurose & Ross (2021): Explains TCP reliability mechanisms.
- Tanenbaum & Wetherall (2021): Discusses layered error control.
- Stallings (2020): Covers CRC and retransmission methods.
- Cisco Networking Academy: Explains reliable communication protocols.

Q5. Web Browsing Using OSI Layers

When a user opens a website, data passes through all seven OSI layers. The Application Layer uses HTTP/HTTPS to request the webpage. The Presentation Layer encrypts and formats data using SSL/TLS. The Session Layer establishes and manages the communication session.

The Transport Layer uses TCP to provide reliable communication. The Network Layer routes packets using IP addresses. The Data Link Layer transmits frames using MAC addresses, while the Physical Layer sends the data as electrical, optical, or wireless signals.

If any layer fails, communication is interrupted. For example, if the Transport Layer fails, TCP cannot establish a reliable connection, preventing the webpage from loading.

Q5. Web Browsing Using OSI Layers

OSI Layer		Role in Web Browsing	Failure in One Layer Affects All
7	 Application	Browser (HTTP/HTTPS request) Requesting web page, receiving response	! Web page not displayed. User cannot access the website.
6	 Presentation	SSL/TLS encryption, data format (HTML, CSS, JS) Encrypts data, converts to common format	! Cannot decrypt / wrong format. Page appears garbled or not loaded.
5	 Session	Session establishment (TCP session) Manages sessions between browser and server	! Session fails / disconnects. User gets logged out or connection reset.
4	 Transport	TCP (reliable delivery, ports) Ensures complete delivery using segments	! Connection fails / data loss. Page loading stops or is incomplete.
3	 Network	IP routing to server Finds best path and routes packets	! No routing / destination unreachable. Packets cannot reach the server.
2	 Data Link	MAC addressing, framing, error detection Delivers frames on the local network	! Frame loss / retransmissions. Slow connection or frequent timeouts.
1	 Physical	Transmit bits over media (Wi-Fi / Ethernet) Sends raw bits as electrical / optical signals	! No signal / no transmission. No data can be sent or received.

Analysis

- Web browsing depends on all OSI layers working together in sequence.
- If any one layer fails, the data cannot move forward or be properly processed, breaking the entire communication.
- Example: If the Network layer fails (no routing), Transport, Session and Application layers cannot function.
- Therefore, all layers are essential for successful and reliable web browsing.

Concept Map Architecture (Q5)

Source Concept (Node A)	Linking Relationship	Target Concept (Node B)
User	Requests	Web Browser
Web Browser	Uses	Application Layer
Application Layer	Sends data through	Presentation Layer
Presentation Layer	Passes data to	Session Layer
Session Layer	Establishes	Communication Session
Transport Layer	Provides	Reliable Delivery
Network Layer	Routes	Packets
Data Link Layer	Delivers	Frames
Physical Layer	Physical Layer	Bits
Successful Transmission	Displays	Web Page

Literature Insights & Evidence (Q5)

- Forouzan (2022): Explains OSI communication in web applications.
- Kurose & Ross (2021): Discusses HTTP and TCP communication.
- Tanenbaum & Wetherall (2021): Describes layered communication.
- Stallings (2020): Explains browser-server communication.
- Cisco Networking Academy: Demonstrates web communication using the OSI model.

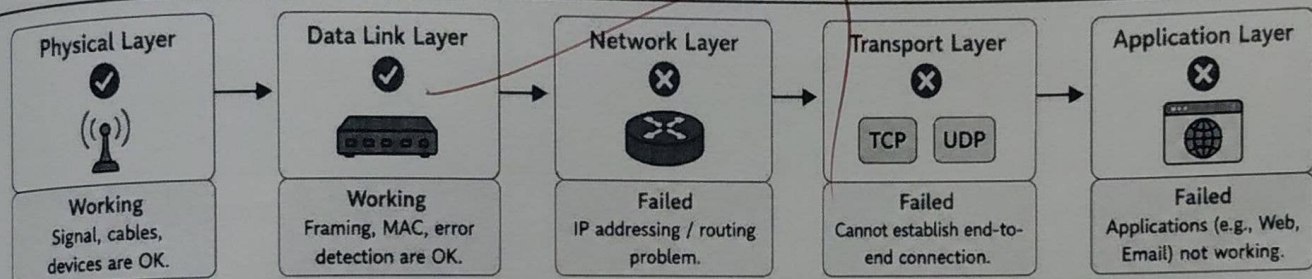
Q6. Troubleshooting Using the OSI Model

The given concept map shows that the Physical Layer and Data Link Layer are functioning correctly, while the Network Layer, Transport Layer, and Application Layer have failed.

Since the Network Layer is responsible for routing packets, its failure prevents packets from reaching the destination network. As a result, the Transport Layer cannot establish end-to-end communication, and the Application Layer cannot deliver services such as web browsing, email, or file transfer.

The OSI model simplifies troubleshooting by identifying the exact layer where communication fails. Network engineers can isolate problems efficiently by checking each layer one at a time.

Q6. Network Issue and Troubleshooting Using OSI Concept Map



Analysis of the Problem

The failure starts at the Network Layer. Because the data cannot be routed to the destination, the upper layers (Transport and Application) also fail.

Effect on Communication

End-to-end communication is not possible.

How Concept Map Helps in Troubleshooting

- Identify Working Layers**
Physical and Data Link layers are working (✓). So the issue is above these layers.
- Locate Point of Failure**
Network Layer shows failure (X). Start investigating here.
- Check Common Causes (Network Layer)**
 - Incorrect IP address / subnet mask
 - Default gateway misconfigured
 - Routing table error / no route to host
- Verify and Fix**
 - Verify IP settings, ping default gateway
 - Check routing table, gateway, firewall
- Re-test Communication**
Once Network layer is fixed, upper layers (Transport, Application) will start working.

Troubleshooting Steps (Summary)

- 1 Ensure Physical connections are OK.
- 2 Confirm Data Link layer (frames, MAC) is working.
- 3 Check IP configuration, routing and connectivity.
- 4 Verify Transport layer (TCP/UDP ports, firewall).
- 5 Check Application (services, URLs, DNS).
- 6 Fix the issue at the failed layer.
- 7 Recheck end-to-end communication.

Takeaway

The concept map clearly shows the flow of communication and helps to quickly locate the layer where failure occurs. Fixing the lower failed layer restores the functioning of all upper layers.

Concept Map Architecture (Q6)

Source Concept (Node A)	Linking Relationship	Target Concept (Node B)
Physical Layer	Successfully Transmits	Bits
Data Link Layer	Successfully Delivers	Frames
Network Layer	Fails to	Route Packets
Routing Failure	Prevents	Transport Layer Communication
Transport Layer Failure	Prevents	Application Layer Services
Application Failure	Results in	Communication Failure
OSI Model	Helps	Network Troubleshooting

Literature Insights & Evidence (Q6)

- Forouzan (2022): Explains troubleshooting using the OSI model.
- Kurose & Ross (2021): Describes layered fault isolation.
- Tanenbaum & Wetherall (2021): Discusses network diagnostics.
- Stallings (2020): Explains troubleshooting techniques for communication networks.
- Cisco Networking Academy: Demonstrates practical OSI troubleshooting methods.

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